

SHAIFALI GARG

Analytical Thinker | Quantitative Researcher | BI Manager

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PROFESSIONAL SUMMARY

Analytical thinker with strong mathematical foundations and ten years tackling complex problems across quantitative disciplines, from supply chain optimization to predictive modeling. Cross-disciplinary expertise grounded in business analytics. Proven ability to rapidly learn new domains and apply rigorous analytical frameworks. Strong communicator translating complex quantitative concepts for interdisciplinary teams. Curiosity-driven researcher committed to leveraging analytical expertise to build impactful AI systems.

CORE COMPETENCIES

Math Modeling, Statistical Analysis, Problem Solving, Quantitative Research, Facilities Services Management, Predictive Analytics, Machine Learning, Rapid Learning, Hypothesis Testing, Regression Analysis, Time Series, Neural Networks

WORK EXPERIENCE

Manager, Business Intelligence and Reporting

2022 - Present

Power Department, Walmart Inc, Bentonville, USA

- Created new processes and workflows in Phoenix (Quickbase) that help run the business and finance across all POs, maintaining financial changes in data directly from SAP.
- Lead cross-disciplinary teams, fostering collaborative problem-solving and a continuous learning culture.
- Translate complex quantitative insights into clear, actionable recommendations for executive leadership.
- Design advanced analytical solutions leveraging statistical methods, predictive modeling, and ML algorithms.
- Demonstrate rapid learning agility by mastering new domains, tools, and methodologies.
- Manage complex BI and reporting solutions by researching potential solutions and making recommendations to meet end-user objectives; updating business data; creating visual prototypes and design documentation; monitoring solutions and troubleshooting issues; completing dashboard and reporting specifications; and creating reports and visualizations to effectively communicate results.
- Manage the implementation of BI solution roadmaps by developing written and oral presentations for diverse audiences (management, customers, suppliers, BI and analytics staff); ensuring solution testing and validation; ensuring requirements are met to support implementations; overseeing upgrades and release schedules for BI and database products; reviewing the impact of changes before and after implementation; and communicating project results and updates to senior leadership.
- Provide supervision and development opportunities for associates by hiring and training on modern tools such as Code Puppy and Copilot; mentoring; assigning duties; providing recognition; and ensuring diversity awareness.
- Create and revise complex BI and reporting solutions by identifying business requirements and validating gathered information; collaborating with subject matter experts, data experts, and other BI and analytics experts; coordinating the evaluation of customer and supplier solutions; making recommendations to meet end-user objectives; designing and conducting analysis and reporting solutions; writing detailed dashboard and reporting specifications; documenting and revising data collection processes; and creating reports and visualizations to effectively communicate results.
- Understand current customer BI and reporting needs by managing production of scheduled reports; managing reporting databases to ensure accuracy and completeness of data; maintaining materials for state-of-the-art user interfaces that enable business users to interpret trends and exceptions; providing guidance and direction to stakeholders; and identifying gaps and opportunities in BI and reporting solutions.

Senior Data Analyst

2022

Supply Chain, Walmart Inc, Bentonville, USA

- Developed 20+ dashboards translating complex data patterns into clear visualizations for diverse stakeholders.
- Applied time series forecasting models for transportation demand, balancing complexity with applicability.
- Elevated customer satisfaction from 60% to 95% through rigorous root cause analysis and systematic frameworks.
- Tackled inventory optimization for 1,500+ SKUs using mathematical modeling and constraint optimization.

Business Analyst Intern

2022

Aezion Inc, Dallas, USA

- Applied quantitative risk analysis to \$2B programs using Monte Carlo simulations.
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Achieved 20% supply chain cost reduction through analytical insights.

Research Analyst

2021

Southern Methodist University, Dallas, USA

- Developed mathematical models in Python quantifying marketing impact on consumer behavior.
- Built neural network models predicting performing arts attendance during COVID-19 unprecedented conditions.
- Created an interactive web application (RShiny & SQL) enabling exploration of complex predictive models.

R&D Quality Assurance Manager

2020 - 2021

Dassault Systemes Solutions Lab, Pune, India

- Led a team of 7 automating analytical workflows (Power BI), achieving 10% revenue increase.
- Defined 100+ KPIs for 6 product launches, establishing rigorous quantitative measurement frameworks.
- Created dashboards translating complex clinical trial data into accessible insights for medical stakeholders.

Senior Consultant

2019 - 2020

Publicis Sapient, Gurgaon, India

- Performed EDA using advanced statistical techniques (Q-Q plots, correlation analysis).
- Conducted rigorous root cause analysis improving software quality.

QA Analyst

2018 - 2019

Capgemini, Pune, India

- Led an analytical team developing quantitative models for media supply chain optimization.
- Performed EDA on financial loan data, uncovering behavioral patterns and risk factors.

Senior Data Analyst

2014 - 2018

Infosys Ltd, Pune, India

- Applied hypothesis testing, regression, and correlation for strategic business decisions.
- Performed linear programming and sensitivity analysis for resource optimization.

ANALYTICAL PROJECTS

Phoenix (Quickbase) Finance Platform: Designed new processes and workflows in Phoenix (Quickbase) to run business and finance operations across all POs, maintaining financial changes in data sourced directly from SAP.

Code Puppy Diaries (Live): A living project exploring AI-assisted development with Code Puppy - experiments, automations, and learnings, versioned and shared on Walmart Enterprise GitHub (gecgithub01.walmart.com/s0g0em7/Code-Puppy-Diaries-live).

EOL Electrical Analysis (Multi-Agent): Automated end-of-life electrical analysis - picks stores one by one, finds the latest diagram model from a shared folder, downloads it, converts it into a readable Excel form, and pushes the data into Phoenix (Quickbase), all orchestrated by 4 parallel Quickbase agents.

Power Automate Ticketing Bot: Built an automation bot in Power Automate that copies comments from an email and pastes them into the DRT associated with a ticket, saving roughly 7 hours every day.

Statistical Decision Modeling: Applied hypothesis testing, multiple regression, and correlation for a multi-million dollar acquisition evaluation (Sleep Cool Mattress).

Consumer Behavior Analysis: Analyzed large-scale Six Flags data identifying behavioral patterns; built interactive Tableau dashboards.

Optimization Modeling: Performed linear programming & sensitivity analysis for LEGO production planning and resource allocation.

EDUCATION

MS in Business Analytics (GPA: 3.7/4.0)

2021 - 2022

Southern Methodist University, Dallas, USA

- Coursework: Statistical Analysis, ML, Predictive Modeling, Decision Optimization, Time Series.

BS in Computer Science & Engineering

2010 - 2014

Lingaya's University, India

- Foundation: Algorithms, Data Structures, Computational Math, Database Systems.

TECHNICAL SKILLS & METHODOLOGIES

Programming & Tools: Python (NumPy, Pandas, Scikit-learn), R (Statistical Analysis), SQL, Tableau, Power BI, GCP BigQuery, Alteryx, Dataiku, Excel (Advanced Analytics), Code Puppy, Git, AI Innovation Lab, Wibey, VS Code

Analytical Methods: Hypothesis Testing, Regression (Linear, Logistic, Multiple), Time Series (ARIMA, Prophet), Neural Networks, Decision Trees, Clustering, Monte Carlo Simulation, Linear Programming, Sensitivity Analysis, Root Cause Analysis, A/B Testing, EDA

Domains: Supply Chain, Healthcare, Finance, Social Media, Performing Arts, Retail, Facilities Services, Walmart Power

CERTIFICATIONS & COMMITMENT

- Certifications: Dataiku Core Designer | Bloomberg Market Concepts | Python Data Science Toolkit (DataCamp) | Data Analytics in R | Lean Six Sigma Yellow Belt (AIGPE)
- Continuous Learning: Rapidly mastered new platforms (Dataiku, Alteryx), demonstrating intellectual curiosity and adaptability.
- Cross-Disciplinary Impact: Applied expertise across supply chain, healthcare, finance, social media, performing arts, and facilities services.
- Collaborative Excellence: Effective communication with technical teams, business stakeholders, and academic researchers.